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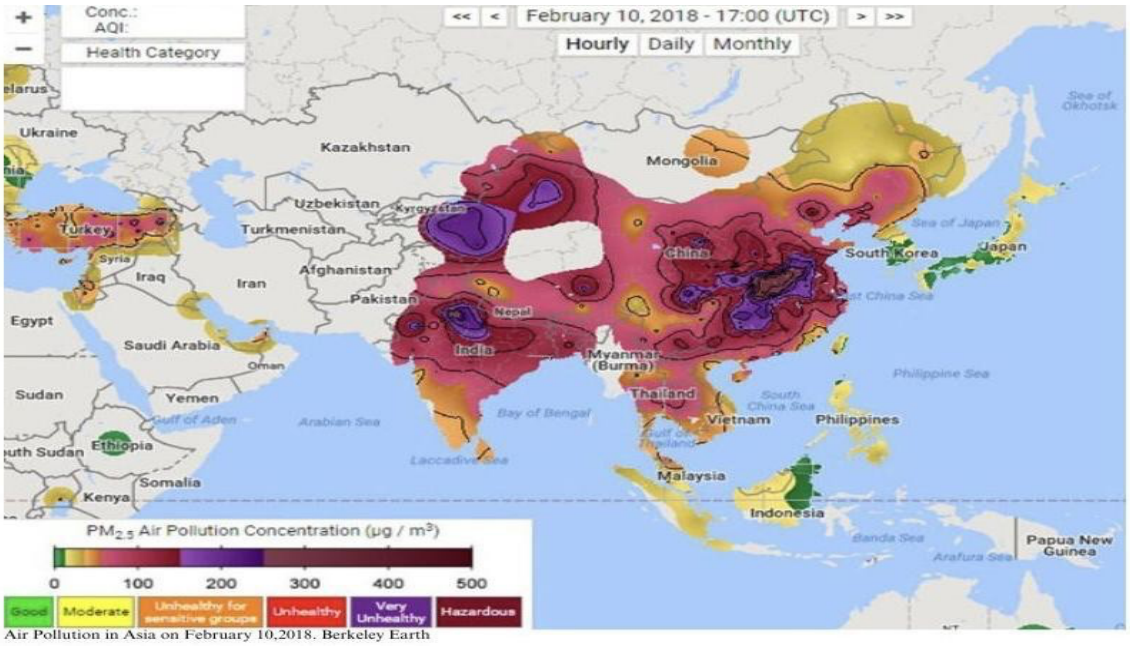
SHARED POWER, SHARED PROSPERITY: Advancing Renewable Energy Through Regional Collaboration in South Asia

Executive Summary:

The transition to renewable energy is critical for South Asia **to balance its economic boom with sustainable development practices**. Yet, significant barriers exist, especially inadequate infrastructure and challenges in regional collaboration. This policy memo outlines actionable policy measures to strengthen regional integration by enhancing **the role of MDBs in establishing a regional power market.** These recommendations will enable South Asia to decarbonize rapidly, strengthen regional security, and meet national and regional priorities.

South Asia's Energy Crisis and Climate Vulnerability:

South Asia is central to the transition to global green energy. Home to more than one-quarter of the world's population, the **region is expected to grow by an additional 600 million people over the next 30 years.** (Fruman & Gupta, 2023). With both the population and economies growing rapidly, coupled with surging demand from manufacturing sectors, South Asia's energy demand is the fastest growing in the world. **Two-thirds of the energy used in South Asia is imported, and 80% of energy production comes from fossil fuels.** (Fruman & Gupta,2023). The inability to scale renewable energy transitions swiftly has exacerbated fossil fuel consumption, increasing carbon dioxide emissions. The consequences are increasingly dire, with severe air quality concerns and flooding in countries like Bangladesh and Indonesia affecting millions. Geographically and socio-economically vulnerable to climate disasters, South Asia's shift to clean energy use is critical to mitigate the adverse impacts of global warming. International climate commitments, such as the Paris Agreement and, recently, the COP28, convened in the UAE, underscore the urgency of accelerating the transition to renewable energy and diversifying energy portfolios towards renewable production. While South Asia is grappling with significant challenges, it also holds **vast potential for renewable energy sources, including wind, hydropower, and solar.**



Air Pollution in Asia on February 10,2018. Berkeley Earth

Channeling these resources can:

1. Address gaps in energy demand
2. Foster sustainable development
3. Enhance energy security in the region.

Many countries in South Asia**,** however, face severe infrastructural limitations and require modern grid infrastructure to support this transition. Massive financing is required for South Asia's transition from both the public and private sectors. (Laxton, Ali, & McGrory, 2024). Since MDBs are at the intersection of the public and private sectors, they can be major players in:

1. De-risking investments and attracting private investment for such projects
2. Working closely with National Governments and helping shape aligned energy policies and regulatory frameworks. (EnviroSolutions & Consulting, 2024)
3. Facilitating regional collaboration.

MDBs have already made significant strides in facilitating this transition, including the **ETM and SARI/EI initiatives.** The Energy Transition Mechanism (ETM) was piloted in Indonesia, Vietnam, and the Philippines and brings together governments, investors, and other stakeholders to accelerate the transition away from coal and scale up investments in renewable energy financing. SARI/EI is one of the largest energy projects in the region, funded by USAID, that works to establish a regional electricity market. (Huda & McDonald, 2016). Through technical assistance and innovative financing mechanisms, **ADB can ensure that South Asia not only transitions to clean energy but does it equitably.**

Energy Demand and Regional Challenges:

By examining the role of **economic, social, and geopolitical factors**, the findings illustrate the barriers to clean energy transition and opportunities where MDBs can strategically intervene:

1) Infrastructural Limitations:

The affordability of the technology and infrastructure required to transition to renewable energy remains a pressing challenge for the South Asian region. A report from the International Energy Agency (IEA) estimates that Southeast Asia requires $190 billion annually to reach its climate goals by 2030. To channel renewable energy, grids need to be modernized with battery storage and power management systems capable of matching demand and supply. (Laxton et al., 2024). Grid modernization projects also require private investment, which is difficult to obtain as grids are publicly owned by the state. Upgrading a grid across different regions also requires obtaining permissions, which is typically a lengthy process. This is where MDBs can come in to overcome these complexities and undertake projects that upgrade existing grids.
For instance, the ADB has designed projects to improve infrastructure capabilities and enhance Nepal's electricity access. (Laxton, et al.2024)

2) Financing Gaps:

There is a lack of flexibility and subsidy mechanisms for cross-border energy projects. Furthermore, due to the Asian Bank-dominated financial system, there is a lack of a welldeveloped market and limited venture capital availability. (Peimani, 2018). Green energy technologies are also relatively new and require substantial initial investments; therefore, banks consider them risky to finance. The private sector is unlikely to invest in such projects for similar reasons, and investors are wary of losses from currency fluctuations.

3) Geopolitical Tensions:

Impediments caused by political conflicts continue to hinder progress, including persistent disputes over borders, historical conflicts, and nationalistic policies that have resulted in low levels of integration in the region. (Huda & McDonald, 2016). Although efforts have been made to remedy this, such as the establishment of the South Asian Association for Regional Cooperation (SAARC). This organization, however, has achieved little success in facilitating trade and connectivity. South Asia's regionalism is shaped by India's relationships with its neighbors, who often view it as a dominant yet unreliable partner, creating barriers to trust and collaboration. (Huda & McDonald, 2016). This issue is also exacerbated by the Indo-China competition as China's investments in Sri Lanka and Nepal pose challenges to building multilateral ties, and there is limited cooperation on cross-border infrastructure projects such as the stalled BCIM economic corridor.

Recommendations:

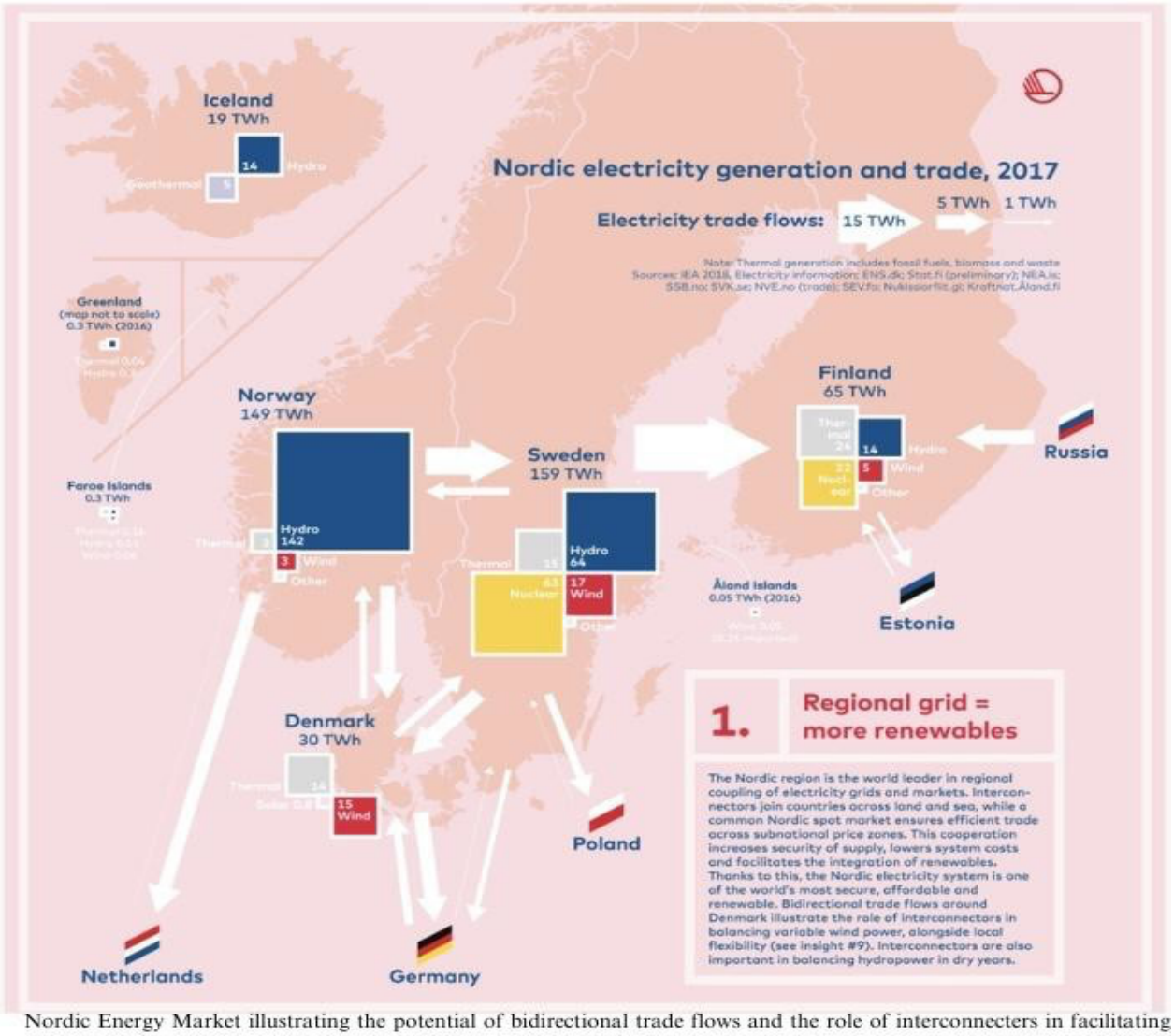
The recommendations outlined below discuss the significance of a regional power market and strategic approaches that MDBs such as ADB must adopt to bring it to reality. These recommendations include **risk-mitigating strategies, raising political will, innovative financing, and enhancing grid infrastructure.**

Establishing a regional electricity market:

A regional electricity market promotes power security, enhances cross-border energy trade, and enables cost reductions associated with green energy economies of scale. MDBs can provide concessional financing or grants to build cross-border transmission lines. They can also serve as neutral parties to bring together stakeholders from multiple countries, acting as mediators for power-sharing agreements and tariff regulation. (Fruman & Gupta, 2023).

Key lessons can be learned from the NORDIC power market that provides a transparent and efficient platform to exchange electricity and boost renewable energy production. The region now has the world's most harmonized cross-border power market. In the context of South Asia, MDBs must focus on enabling governments and other regional organizations to set ambitious targets and policies that strengthen political commitment. (Mundaca, Dalhammar, & Harnesk, 2013).

According to a World Bank report, a regional power market in South Asia can lead to up to $17.4 billion in economic benefits between 2023 and 2035. This will pave the way for a carbon-free future for South Asia. (Fruman & Gupta, 2023).



Nordic Energy Market illustrating the potential of bidirectional trade flows and the role of interconnecters in facilitating trade.
Nordic Energy Research

Core actions under this recommendation can look like:

- <u>Partnering with national governments to raise the political importance of regional organizations:</u>
  A major reason for the ineffectiveness of regional organizations such as SAARC in initiating green energy projects is the lack of political will attached to them. **Regional goals set at the SAARC energy center must be internalized and absorbed at the national level by member countries** (Huda & McDonald, 2016). This adoption will be transformative, and SAARC can be a space that facilitates high-level green energy projects. By leveraging their advocacy and convening power, MDBs can organize meetings among stakeholders across SAARC nations and provide technical support, thereby strengthening their institutional capacity.
  By adopting this approach, MDBs can empower South Asian nations to align their energy policies with existing climate agreements and build multilateral partnerships that are lacking in the region. In the case of SAARC nations, this could look like MDBs facilitating talks that can lead to India providing land access to Bangladesh to import electricity from Nepal and Bhutan.

- Conducting feasibility studies on de-risking energy markets:

  Historically, the private sector has failed to participate in green energy development due to concerns about the bankability of such projects (Peimani, 2016). The regulatory environment may also be risky due to weak legal frameworks or contradictory policies.
  De-risking in this context means reducing or sharing the risks associated with climate investments. By conducting feasibility studies, MDBs can identify and address potential risks and shape projects in a manner that attracts investments. (Choi, Zhou, & Laxton, 2022). For instance, the Global Infrastructure Facility provides funding and technical assistance to governments and multilateral development banks to help them select, structure, and bring to market high-quality bankable infrastructure projects.

- Developing transparent and enforceable non-compliance frameworks:

  The Nordic experience suggests **developing strong renewable energy policy frameworks** that address non-compliance and ensure effective enforcement. The penalties for non-compliance involve enforcement of ceiling prices for tradable green certificates (Mundaca et al. 2013). In South Asia, given its fragmented nature and political competition, MDBs, such as ADB, must leverage their roles to provide technical assistance by developing templates for regional electricity market frameworks. This will include **standardizing energy contracts** within the region and developing mechanisms to set **cross-border electricity pricing.**

- Strengthening grid infrastructure:

  Upgrading grid infrastructure is essential to establishing a regional power market in South Asia, as it enables better integration of diverse renewable energy sources, greater reliability during extreme weather events, and more efficient cross-border electricity trading. (Chaudhury, Bose, Samanta, Anbumozhi, 2023). The MDBs play a critical role in strengthening infrastructure by providing concessional loans that have low interest rates, and credit guarantees that reduce risk for private investors. **The current grid infrastructure in South Asia is not equipped to handle the flow of electricity trade within the region.** Therefore, MDBS must provide financial and technical support to implement advanced grid technologies that ensure a reliable power supply and support regional integration in the power sector.

Call to Action: MDBs as Catalysts for Change

South Asia stands at a pivotal moment, where the transition to renewable energy presents an unparalleled opportunity to drive sustainable development while meeting its surging energy demands. A dedicated multilateral platform, spearheaded by Multilateral Development Banks (MDBs), is crucial to unite stakeholders in addressing the multifaceted challenges associated with green energy transition. (Fruman & Gupta, 2023).
Given its position as **one of the world's most climate-vulnerable regions, immediate and coordinated action is imperative.** MDBs are well-positioned to lead this transformation by developing targeted interventions to build resilient energy systems that ensure compatibility between growth and development targets.

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